

Foundations of Audio Signal Processing

Exercise sheet 4

To be uploaded in eCampus till: 14-11-2025 22:00 (strict deadline)

Exercise 4.1

[3 + 3 + 3 = 9 points]

Indicate whether the following mappings $d: \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ define a metric on $\mathbb{C}$. Prove your statement.

- (a) $d(x, y) := |x - y|$
- (b) $d(x, y) := |x| \cdot |y|$
- (c) $d(x, y) := \begin{cases} 1, & \text{if } x \neq y, \\ 0 & \text{else.} \end{cases}$

Exercise 4.2

[3 + 4 = 7 points]

A function $f: \mathbb{Z} \rightarrow \mathbb{C}$ is called *periodic*, if there exists a $p \in \mathbb{N}$, such that $f(n + p) = f(n)$ for all $n \in \mathbb{Z}$. For $\omega \in [0, 1)$ we define the following discrete frequency signal: $f_\omega: \mathbb{Z} \rightarrow \mathbb{C}$ as $f_\omega(n) := e^{2\pi i \omega n}$.

- (a) Draw figures of $f_{\frac{1}{2}}$, $f_{\frac{1}{5}}$, $f_{\frac{1}{6}}$ in the complex plane.
- (b) Prove that: f_ω periodic $\Leftrightarrow \omega \in \mathbb{Q}$.

Exercise 4.3

[4 points]

Prove the theorem of Pythagoras:

If $x_1, \dots, x_n$ are pairwise orthogonal elements of the Hilbert space V , then the energy of the sum of the x_i is the same as the sum of the single energies:

$$\left\| \sum_{j=1}^n x_j \right\|^2 = \sum_{j=1}^n \|x_j\|^2.$$

Explain the steps of your proof.